

習題 2.17

$$u(x, y, t) = \frac{1}{2\pi a} \int_0^t \iint_{|\vec{y}-\vec{x}| \leq a(t-\tau)} \frac{f(\tau) \delta(y_1) \delta(y_2)}{\sqrt{a^2(t-\tau)^2 - |\vec{y}-\vec{x}|^2}} dy_1 dy_2 d\tau$$

若 $|\vec{x}| \geq at$, 則 $f(\tau) \delta(y_1) \delta(y_2) = 0$

$$\Rightarrow u(x, y, t) = 0$$

$$\text{若 } |\vec{x}| \leq at, \text{ 則 } u(x, y, t) = \frac{1}{2\pi a} \int_0^t \frac{f(\tau)}{\sqrt{a^2(t-\tau)^2 - (x^2+y^2)}} d\tau$$

習題 2.18

$$\partial_t u(x, t) - a^2 \partial_{xx} u(x, t) = f(x, t) \Leftrightarrow \partial_t u(-x, t) - a^2 \partial_{xx} u(-x, t) = f(-x, t) = f(x, t)$$

$$u(x, 0) = \varphi(x) \Leftrightarrow u(-x, 0) = \varphi(-x) = \varphi(x)$$

因此 $u(-x, t)$ 也是一維熱傳導方程初值問題的解。

由解的唯一性知, $u(-x, t) = u(x, t)$

習題 2.19

$$(i) \begin{cases} \partial_t u - 4\partial_{xx} u - \partial_x u - 3u = f(x, t) \\ u(x, 0) = \varphi(x) \end{cases} \Rightarrow \begin{cases} \partial_t U(\omega, t) - (4\omega^2 + \omega + 3)U(\omega, t) = F(\omega, t) \\ U(\omega, 0) = \Psi(\omega) \end{cases}$$

$$U(\omega, t) e^{-(4\omega^2 + \omega + 3)t} = \int_0^t F(\omega, \tau) e^{-(4\omega^2 + \omega + 3)\tau} d\tau + \Psi(\omega)$$

$$U(\omega, t) = \int_0^t F(\omega, \tau) e^{(4\omega^2 + \omega + 3)(t-\tau)} d\tau + \Psi(\omega)$$

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \left(\int_0^t F(\omega, \tau) e^{(4\omega^2 + \omega + 3)(t-\tau)} d\tau \right) e^{i\omega x} d\omega + \varphi(x)$$

$$(ii) \begin{cases} \partial_t u - 4\partial_{xx} u - \partial_x u - 3u = \hat{f}(x, t) \\ u(x, 0) = \varphi(x) \end{cases} \Rightarrow \begin{cases} \partial_t U + (4\omega^2 - i\omega - 3)U = \hat{f}(\omega, t) \\ U(\omega, 0) = \Psi(\omega) \end{cases}$$

$$U(\omega, t) e^{(4\omega^2 - i\omega - 3)t} = \int_0^t \hat{f}(\omega, \tau) e^{(4\omega^2 - i\omega - 3)\tau} d\tau + \Psi(\omega)$$

$$U(\omega, t) = \int_0^t \hat{f}(\omega, \tau) e^{(4\omega^2 - i\omega - 3)(\tau-t)} d\tau + \Psi(\omega) e^{-(4\omega^2 - i\omega - 3)t}$$

$$u(x, t) = \frac{1}{\sqrt{2\pi}} \left[\int_{-\infty}^{+\infty} \left[\int_0^t \hat{f}(\omega, \tau) e^{(4\omega^2 - i\omega - 3)(\tau-t)} d\tau \right] e^{i\omega x} d\omega + \int_{-\infty}^{+\infty} \Psi(\omega) e^{-(4\omega^2 - i\omega - 3)t} e^{i\omega x} d\omega \right]$$

$$= \int_0^t \frac{1}{\sqrt{2\pi}} \left(\hat{f}(x, \tau) * e^{-(4\omega^2 - i\omega - 3)(t-\tau)} \right) e^{i\omega x} d\omega +$$

$$= \int_0^t \frac{1}{\sqrt{2\pi}} \left(f(x, \tau) * F^{-1}[e^{(4\omega^2 - i\omega - 3)(t-\tau)}] \right) d\tau + \frac{1}{\sqrt{2\pi}} \varphi(x) * F^{-1}[e^{-(4\omega^2 - i\omega - 3)t}]$$

$$F^{-1}[e^{-(4\omega^2 - i\omega - 1)t}] = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-4\omega^2 t + i\omega t + 3t} e^{i\omega x} d\omega$$

$$= \frac{e^{3t}}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} e^{-4\omega^2 t} e^{i\omega(x+t)} d\omega = \frac{e^{3t}}{\sqrt{2\pi}} F^{-1}[e^{-4\omega^2 t}](x+t)$$

$$F[e^{-A\omega^2}] = \sqrt{\frac{\pi}{A}} e^{-\frac{x^2}{4A}} \Rightarrow F^{-1}[e^{-\frac{\omega^2}{4A}}] = \sqrt{\frac{A}{\pi}} e^{-Ax^2}$$

$$\text{Für } 4t = \frac{1}{4A} \Leftrightarrow A = \frac{1}{16t}, \text{ f\"ur } F^{-1}[e^{-(4\omega^2 - i\omega - 1)t}] = \frac{e^{3t}}{\sqrt{16\pi t}} e^{-\frac{1}{16t}(x+t)^2}$$

$$= \frac{e^{3t}}{\sqrt{8t}} e^{-\frac{1}{16t}(x+t)^2}$$

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \left[\int_0^t f(x,\tau) * \frac{e^{3(t-\tau)}}{\sqrt{8(t-\tau)}} e^{-\frac{1}{16(t-\tau)}(x+\tau-\tau)^2} d\tau + \varphi(x) * \frac{e^{3t}}{\sqrt{8t}} e^{-\frac{1}{16t}(x+t)^2} \right]$$

$$(2) \begin{cases} \partial_{tt}u + 2\partial_tu - \partial_{xx}u + u = 0 \\ u(x,0) = 0, \partial_tu(x,0) = x \end{cases} \Rightarrow \begin{cases} \partial_{tt}\hat{u} + 2\partial_t\hat{u} + (\omega^2 + 1)\hat{u} = 0 \\ \hat{u}(\omega,0) = 0, \partial_t\hat{u}(\omega,0) = \lambda \cdot \sqrt{2\pi} \frac{d}{d\omega} \delta(\omega) \end{cases}$$

$$\lambda^2 + 2\lambda + 1 + \omega^2 = 0$$

$$\Rightarrow \lambda = -1 \pm \omega i \Rightarrow \hat{u} = A e^{-1+\omega i} t + B e^{-1-\omega i} t$$

$$\Rightarrow \hat{u}(\omega, t) = A(\omega) e^{(-1+\omega i)t} + B(\omega) e^{(-1-\omega i)t}$$

$$\begin{cases} A(\omega) + B(\omega) = 0 \\ A(\omega)(-1+\omega i) + B(\omega)(-1-\omega i) = \sqrt{2\pi} \lambda \delta'(\omega) \end{cases}$$

$$\Rightarrow 2\omega i B(\omega) = -\sqrt{2\pi} \lambda \delta'(\omega)$$

$$B(\omega) = -\sqrt{\frac{\pi}{2}} \frac{\delta'(\omega)}{\omega} = -A(\omega)$$

$$\hat{u}(\omega, t) = \sqrt{\frac{\pi}{2}} \frac{\delta'(\omega)}{\omega} e^{-t} \cdot 2i \sin \omega t$$

$$u(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{+\infty} \sqrt{\frac{\pi}{2}} e^{-t} \hat{x} \frac{\sin \omega t}{\omega} e^{i\omega x} d\omega$$

$$= e^{-t} F^{-1}\left[\hat{x} \frac{\sin \omega t}{\omega}\right] = \frac{1}{\sqrt{2\pi}} e^{-t} x * F^{-1}\left[\frac{\sin \omega t}{\omega}\right]$$

$$= \frac{1}{\sqrt{2\pi}} e^{-t} x * \left(\int_0^1, |x| \leq t \right) = e^{-t} \frac{1}{2} \int_{x-t}^{x+t} d\xi = \frac{e^{-t}}{2} \cdot 2xt = xte^{-t}$$

$$(3) \begin{cases} \partial_{tt} u - \partial_{xx} u = t \sin x \\ u(x, 0) = 0, \quad \partial_t u(x, 0) = \sin x \end{cases}$$

$$\Rightarrow \begin{cases} \partial_{tt} \hat{u} + \omega^2 \hat{u} = \frac{t}{2\lambda} (\sqrt{2\pi} \delta(\omega-1) - \sqrt{2\pi} \delta(\omega+1)) \\ \hat{u}(\omega, 0) = 0, \quad \partial_t \hat{u}(\omega, 0) = \frac{1}{2\lambda} (\sqrt{2\pi} \delta(\omega-1) - \sqrt{2\pi} \delta(\omega+1)) \end{cases}$$

$$\hat{u}(\omega, t) = \frac{\sqrt{2\pi} t}{2\lambda \omega} (\delta(\omega-1) - \delta(\omega+1)) + C_1 e^{-i\omega t} + C_2 e^{i\omega t}$$

$$\Rightarrow \begin{cases} C_1 + C_2 = 0 \\ (C_2 - C_1) = 0 \end{cases} \Leftrightarrow C_1 = C_2 = 0$$

$$\hat{u}(\omega, t) = \sqrt{\frac{\pi}{2}} \cdot \frac{t}{\lambda \omega} (\delta(\omega-1) - \delta(\omega+1))$$

$$u(x, t) = \frac{1}{2\lambda} \int_{-\infty}^{+\infty} \frac{t}{\lambda \omega} (\delta(\omega-1) - \delta(\omega+1)) e^{i\omega x} d\omega$$

$$= \frac{t}{2\lambda} \left(e^{ix} - \frac{e^{-ix}}{i} \right) = \cancel{\frac{t}{2\lambda}} t \sin x$$

習題 2.20.

設 $\bar{u}(x, p) = L[u(x, y)]_y$

$$\bar{u}(0, p) = \int_0^{+\infty} (y+1)e^{-py} dy = \frac{1}{p} + \frac{1}{p^2}$$

$$L[\partial_y u(x, y)]_y = p \bar{u}(x, p)$$

$$L[\partial_{xy} u(x, y)]_y = p \partial_x \bar{u}(x, p) = \frac{1}{p} \Rightarrow \partial_x \bar{u}(x, p) = \frac{1}{p^2} \Rightarrow \bar{u}(x, p) = \frac{x}{p^2} + C(p)$$

$$\Rightarrow \bar{u}(x, p) = \frac{x+1}{p^2} + \frac{1}{p} \Rightarrow u(x, y) = (x+1)y + 1 = \frac{x}{p^2} + \frac{1}{p} + \frac{1}{p^2}$$

習題 2.21

設 $\forall \vec{x} \in \mathbb{R}^2 \setminus B_\varepsilon$, 取 $\vec{r} = \vec{x} - \vec{\xi}$

~~$$\Delta u_\varepsilon = \frac{\partial^2 u_\varepsilon}{\partial x_1^2} + \frac{\partial^2 u_\varepsilon}{\partial x_2^2}, \quad \frac{\partial u_\varepsilon}{\partial x_1} = \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{\partial r}{\partial x_1} + \frac{\partial u_\varepsilon}{\partial \theta} \cdot \frac{\partial \theta}{\partial x_1}$$~~
~~$$= \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{r}{x_1} + \frac{\partial u_\varepsilon}{\partial \theta} \cdot \left(-\frac{1}{x_2}\right) \quad \left(\frac{\partial u_\varepsilon}{\partial \theta} = 0\right)$$~~

~~$$\frac{\partial^2 u_\varepsilon}{\partial x_1^2} =$$~~

~~$$= \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{r}{x_1} = \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{1}{\cos \theta}$$~~

~~$$\frac{\partial^2 u_\varepsilon}{\partial x_1^2} = \left(\frac{\partial^2 u_\varepsilon}{\partial r^2} \cdot \frac{r^2}{x_1^2} + \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{2r}{x_1} \right) + \frac{\partial u_\varepsilon}{\partial r}$$~~

$$\frac{\partial u_\varepsilon}{\partial x_1} = \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{\partial r}{\partial x_1} = \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{x_1}{r}$$

~~$$\frac{\partial^2 u_\varepsilon}{\partial x_1^2} = \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{\partial}{\partial r} \left(\frac{\partial u_\varepsilon}{\partial r} \cdot \frac{x_1}{r} \right) = \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{\partial^2 u_\varepsilon}{\partial r^2} \cdot \frac{x_1}{r} - \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{x_1}{r^2}$$~~

~~$$\frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{\partial^2 u_\varepsilon}{\partial r^2} = \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{\partial}{\partial r} \left(\frac{\partial u_\varepsilon}{\partial r} \cdot \frac{x_1}{r} \right)$$~~

習題 2.21

$$\Delta u_\varepsilon = \frac{\partial^2 u_\varepsilon}{\partial x_1^2} + \frac{\partial^2 u_\varepsilon}{\partial x_2^2}$$

$$= \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{\partial^2 u_\varepsilon}{\partial r^2}$$

~~$$= \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r}$$~~

$$= \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial u_\varepsilon}{\partial r} \right)$$

由題意可知

$$\frac{\partial u_\varepsilon}{\partial \theta} = 0.$$

$$\frac{\partial u_\varepsilon}{\partial x_1} = \frac{\partial u_\varepsilon}{\partial r} \cdot \frac{x_1}{r}, \quad \frac{\partial^2 u_\varepsilon}{\partial x_1^2} = \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{x_1}{r} \cdot \frac{\partial^2 u_\varepsilon}{\partial x_1 \partial r}$$

$$= \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{x_1}{r} \cdot \frac{\partial}{\partial r} \left(\frac{\partial u_\varepsilon}{\partial r} \cdot \frac{x_1}{r} \right)$$

$$= \frac{1}{r} \frac{\partial u_\varepsilon}{\partial r} + \frac{x_1^2}{r} \cdot \left[\frac{\partial^2 u_\varepsilon}{\partial r^2} \cdot \frac{1}{r} + \frac{\partial u_\varepsilon}{\partial r} \cdot \left(-\frac{1}{r^2}\right) \right]$$

$$= \frac{r^2 - x_1^2}{r^3} \frac{\partial u_\varepsilon}{\partial r} + \frac{x_1^2}{r^2} \frac{\partial^2 u_\varepsilon}{\partial r^2}$$

$$\text{設 } \vec{x} - \vec{\xi} = (r \cos \theta, r \sin \theta) \Rightarrow \begin{cases} \frac{\partial}{\partial r} \left(r \frac{\partial u_\varepsilon}{\partial r} \right) = 0, & r > \varepsilon \\ -2\pi \varepsilon \frac{\partial u_\varepsilon}{\partial r} = 1, & r = \varepsilon \end{cases}$$

$$\Rightarrow r \frac{\partial u_\varepsilon}{\partial r} = -\frac{1}{2\pi} \Rightarrow u_\varepsilon(r, \theta) = -\frac{1}{2\pi} \ln r + C, \quad r \geq \varepsilon. \Leftrightarrow u_\varepsilon(\vec{x}) = \frac{1}{2\pi} \ln \frac{1}{|\vec{x} - \vec{\xi}|} + C$$